Feature-Set Scaling for Kidney Failure Prediction: Benchmarking Eleven Survival Models Against the Kidney Failure Risk Equation on MIMIC-IV

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Abstract

The Kidney Failure Risk Equation (KFRE) predicts progression to end-stage renal disease (ESRD) from as few as four routinely collected variables, yet modern EHR resources and deep survival architectures raise the question of whether more features and more expressive models meaningfully improve on it. We isolate this question on a shared MIMIC-IV-derived cohort of chronic kidney disease patients by benchmarking eleven survival models — Cox, Weibull AFT, DeepSurv, Dynamic-DeepHit, RNN-Surv, Hazard Transformer, LogisticHazard, gradient-boosted survival analysis, random survival forests, FastSurvivalSVM, and the published closed-form KFRE equation itself — across three nested feature-set regimes reconstructing KFRE's own 4- and 8-variable inputs plus a 20-feature heterogeneous stress test. Building KFRE-comparable cohorts required a new creatinine-anchored, bounded-window row-merge design for combining multiple asynchronously-drawn labs into simultaneous covariate snapshots, which we document with its full clinical and statistical justification. Once every model is evaluated on an identical, per-patient basis — a methodological correction that reverses an earlier, row-level-evaluated result in which KFRE and Cox appeared to lead — KFRE's discrimination is matched by most learned models rather than clearly beaten by them, only one deep sequence model (Dynamic-DeepHit) shows a decisive advantage, and nearly every model exceeds a simple eGFR-threshold referral rule's clinical net benefit. We report this as an interim benchmark: two model-behavior findings (a Cox model performing below chance, and Dynamic-DeepHit's uninvestigated architectural advantage) and the pending completion of the twenty-feature scenario and additional cohort repetitions are disclosed explicitly as open questions rather than smoothed over.

Keywords: Survival analysis, chronic kidney disease, Kidney Failure Risk Equation, MIMIC-IV, benchmark

1 Introduction

Chronic kidney disease (CKD) affects over 1 in 7 Americans [1] and can progress to end-stage renal disease (ESRD), which carries a five-year survival rate below 50%. Early identification of patients at high risk of progression is critical for timely nephrology referral, dialysis-access planning, and transplant work-up [2]. The clinical standard for this task is the Kidney Failure Risk Equation (KFRE) [3], a closed-form Cox model built from as few as four routinely collected variables (age, sex, estimated glomerular filtration rate (eGFR), and urine albumin-to-creatinine ratio (uACR)) or, in its extended form, eight variables (adding serum calcium, phosphate, bicarbonate, and albumin). KFRE has been externally validated across 31 cohorts spanning more than 700,000 patients [4] and is now embedded in clinical guidelines [5].

KFRE's minimalism is also its central design question: it was derived under classical regression sample-size constraints, using a handful of variables chosen for clinical availability and interpretability rather than predictive ceiling. Modern large-scale EHR resources and expressive deep survival architectures raise a natural question that KFRE's own literature does not answer: *does adding more labs, and modeling them with more flexible architectures, actually buy additional predictive power for kidney failure risk, or does the four-to-eight variable KFRE feature set already capture most of the recoverable signal?* A companion question is methodological: recent CKD progression models increasingly draw on a much wider EHR feature surface [6] and recommend expanded panels in the most recent clinical guidelines [7], but the benchmarks used to justify this typically evaluate model architecture and feature count as a single, confounded choice.

This paper isolates that confound. Using the same MIMIC-IV-derived CKD cohort construction as our prior MIMIC-CKD benchmark [8], we hold the underlying data source and patient population fixed and vary only the feature set, benchmarking three nested feature-set regimes against 11 survival models each:

- **Four features** — age, sex, eGFR, uACR — an exact reconstruction of KFRE's 4-variable input, letting us benchmark every model directly against the published closed-form equation on its own home turf.
- **Eight features** — the above plus calcium, phosphate, bicarbonate, and serum albumin — KFRE's extended 8-variable input.
- **Twenty features (heterogeneous)** — the 20 most frequently drawn labs for this cohort, each represented as a (value, missingness-flag) pair, following the heterogeneous time-varying representation used in our prior 50-feature benchmark and the panel sizes reported in several recent CKD machine-learning progression models [6]. This scenario has no published KFRE analogue and instead stress-tests whether scaling up the feature count alone improves discrimination.

For the four- and eight-feature scenarios, we additionally implement KFRE itself as a twelfth, non-trained benchmark model — a closed-form risk score computed directly from Tangri et al.'s published coefficients [3, 4] — so that every learned model in this paper is compared against the actual clinical equation it is meant to improve upon, not merely against each other.

Constructing the four- and eight-feature cohorts required a row-merge design not needed by our earlier work: KFRE's inputs are multiple, asynchronously-drawn labs (creatinine, urine albumin/creatinine, and, for the 8-variable form, a chemistry panel) that must be combined into simultaneous covariate snapshots without collapsing a patient's repeated measurements into a single row. We describe this design, and the clinical/statistical precedent behind each of its choices, in Section 2.2.

Our contributions are:

- A cohort-construction methodology for merging multiple asynchronously-drawn labs into KFRE-comparable, patient-level landmark rows from a large heterogeneous EHR source, with each design choice backed by prior clinical/statistical literature (Section 2.2).
- A direct, apples-to-apples benchmark of 11 survival models plus the published KFRE equation across three nested feature-set sizes on the same MIMIC-IV cohort, evaluated with discrimination (C-index, time-dependent AUC), calibration, and decision-curve analysis (Section 3).
- A methodological finding, surfaced during our own debugging process and reported transparently here: evaluating time-varying survival models on their raw per-visit rows rather than one prediction per patient inflates discrimination metrics for several architectures, including KFRE itself — a pitfall we suspect affects other EHR-derived time-to-event benchmarks that reuse a counting-process row format (Section 2.5).
- A candid account of two model-behavior findings that remain open at the time of writing — a Cox model that performs below chance on several feature sets, and one deep architecture (Dynamic-DeepHit) whose margin over every other model is large enough to warrant further scrutiny before being treated as a settled result — reported as limitations rather than smoothed over (Section 4.1).

2 Methods

2.1 Problem Formulation

We frame progression to ESRD as a right-censored time-to-event problem. Each row of covariates $\boldsymbol{x}_i \in \mathbb{R}^d$ is paired with a duration $t_i \in \mathbb{R}^+$ and an event indicator $e_i \in \{0, 1\}$ ($e_i = 1$: ESRD onset observed; $e_i = 0$: right-censored). As in our prior MIMIC-CKD benchmark, patient histories are stored in a counting-process (`start`, `stop`) format so the same row schema supports both single-landmark models (Cox, KFRE) and sequence models that consume a patient's full row history (Dynamic-DeepHit, RNN-Surv). All three feature-set scenarios below share this schema; they differ only in d and in how each row's covariates are populated.

2.2 Cohort

The source population is identical to our prior MIMIC-CKD benchmark: 34,332 MIMIC-IV [8] patients meeting a CKD stage 3–5 or ESRD diagnosis-code criterion, 57.2% male, mean age 67.8 (SD 16.1). What differs in this work is how covariates are assembled into rows for each feature-set scenario.

2.2.1 Four and eight features (KFRE-comparable)

KFRE's inputs are drawn from separate, asynchronously-ordered labs (serum creatinine, a urine albumin/creatinine test, and, for the 8-variable form, a chemistry panel), unlike our prior benchmark's scenarios, which avoid multi-lab merging altogether by encoding one row per single-lab event with missingness flags. Producing real, simultaneous, non-missing covariate rows required a merge design with two open subproblems: (1) whether the row format should still be time-varying given that KFRE itself is a point-in-time equation, and (2) how to handle a patient having multiple draws of the same lab within one admission.

Row format. We keep the time-varying (`start`, `stop`) format used throughout this benchmark family. KFRE's formula is agnostic to row packaging — it can be applied independently to each row, treating every row as its own landmark snapshot in the sense of landmarking methodology for time-varying covariates in survival analysis [9, 10].

Merge rule: creatinine-anchored, bounded nearest-value windows. Each row is anchored on one creatinine measurement (eGFR is computed from it, via the CKD-EPI 2021 equation, at that instant). Other required labs are attached to the anchor by nearest-value matching within an explicit, literature-backed window rather than by unbounded last-observation-carried-forward, which is known to bias sparse EHR-derived time-varying covariates [11]:

- *Calcium, phosphate, bicarbonate, serum albumin* (8-variable scenario only, routinely drawn on the same chemistry panel as creatinine): nearest value within ±24 hours of the anchor. This window follows APACHE II's precedent of treating all labs drawn within a patient's first 24 hours — including serum creatinine — as one physiologic snapshot [12]; no directly analogous 24h-snapshot convention was found for other candidate windows (e.g., KDIGO's 48h AKI change-detection window measures a *trend*, not concurrency, and is not equivalent).
- *uACR* (a separately-ordered urine test): nearest value across the patient's entire history. We initially bounded this match to the same hospital admission, following the resolution Tangri et al. document for one of their own real-world validation cohorts (*"Geisinger: covariates obtained most closely to index date within a past year were included in models"* [4]), but this bound caused 96.5% patient attrition (34,332 → 1,220 for the 4-feature scenario) and skewed the outcome rate from 91.9% to 98.2% ESRD-positive by disproportionately dropping ESRD-negative patients. We loosened the bound to the patient's whole history, trading tighter same-encounter simultaneity for a workable, less-biased cohort size — an accepted, documented tradeoff, not an oversight.

- Any row missing a required lab within its window is *dropped*, not imputed or flagged — unlike our prior benchmark's missingness-flag scenarios, four_features/eight_features are complete-case by construction, matching KFRE's own derivation cohort's eligibility criteria [3].

One row is emitted per qualifying creatinine draw, so a patient monitored multiple times within one admission correctly yields multiple rows, each independently paired with its own nearest in-window value of every other required lab (a sparser lab such as uACR is therefore legitimately reused, at its nearest value, across several rows for the same patient — expected under the landmarking framework above, not a bug).

2.2.2 Twenty features (heterogeneous)

Following our prior benchmark's heterogeneous representation, we select the 20 most frequently measured labs for this cohort and represent each as a (value, missingness-flag) pair, giving a 40-column row, one row per lab event (no cross-lab merge). This scenario intentionally forgoes the four-/eight-feature scenarios' complete-case, KFRE-comparable design in exchange for testing whether feature count alone, at realistic EHR sparsity, adds discriminative power.

2.2.3 Resulting cohorts

Table 1 reports each scenario's final extracted cohort next to the shared source population, following the two-column cohort-flow reporting convention of Major et al. [5] and STROBE item 13(a) [13]. All three scenarios are event-rich by construction (83–92% patient-level ESRD-positive), because the source population is itself drawn from a CKD/ESRD diagnosis-code criterion; we discuss the implications of this for calibration in Section 3.

	Source cohort	**four_features**	**eight_features**	**twenty_features (heterog.)**
Patients	34,332	2,809	1,213	32,601
Records (rows)	74,197	26,829	5,407	8,126,090
% male	57.2	58.3	60.4	57.1
Mean age, yrs (SD)	67.8 (16.1)	63.9 (15.1)	62.1 (15.4)	67.8 (16.1)
Median eGFR (IQR)	N/A	62.1 (36.6–93.0)	61.3	67.5 (45.4–93.6)
Median uACR, mg/g (IQR)	N/A	54.1 (13.0–266.7)	56.0	N/A
Median follow-up, yrs (IQR)	N/A	0.2 (0.0–2.9)	N/A	0.4 (0.0–2.2)
ESRD events, patients (%)	31,542 (91.9)	2,346 (83.5)	1,032 (85.1)	29,815 (91.5)
ESRD rate /1,000 person-yrs	N/A	461.9	N/A	556.2
Train / test patients	—	2,247 / 562	970 / 243	26,080 / 6,521

Table 1 Cohort flow: shared MIMIC-IV source population versus each scenario's final extracted, feature-complete cohort. Splits are 80/20 by patient ID.

Cohort-size adequacy. eight_features' 1,213 patients / 1,032 events is the smallest cohort in this study. Benchmarked against the KFRE literature itself, this is not undersized: it exceeds 9 of the 15 published cohorts (263–12,955 patients) on which the 8-variable KFRE was externally validated [4], and its events-per-variable ratio (1,032 events / 8 variables $\approx$ 129) is roughly 13$\times$ the classical EPV$\geq$10 adequacy floor for stable regression coefficients [14]. This EPV heuristic directly supports the Cox and KFRE benchmarks; it has no established analogue for the more data-hungry deep architectures also benchmarked here, so eight_features remains the scenario most likely to be data-constrained for those models specifically.

2.3 KFRE Baseline

For four_features and eight_features, we add KFRE itself as a twelfth, non-trained benchmark: a closed-form risk score computed directly from Tangri et al.'s published coefficients, requiring no `.fit()` step. Risk at horizon t is $\text{Risk}(t) = 1 - S_0(t)^{\exp(L)}$, using the North American baseline survival constant $S_0(2\text{yr}) = 0.9750$ (this cohort is US-based) [4], with the linear predictor:

$$\begin{aligned} L_4 = &-0.2201\left(\tfrac{\text{age}}{10} - 7.036\right) + 0.2467(\text{male} - 0.5642) \\ &- 0.5567\left(\tfrac{\text{eGFR}}{5} - 7.222\right) + 0.4510\left(\ln(\text{uACR}) - 5.137\right) \end{aligned} \quad (1)$$

for the 4-variable form [3], and

$$\begin{aligned} L_8 = &-0.1992\left(\tfrac{\text{age}}{10} - 7.036\right) + 0.1602(\text{male} - 0.5642) \\ &- 0.4919\left(\tfrac{\text{eGFR}}{5} - 7.222\right) + 0.3364\left(\ln(\text{uACR}) - 5.137\right) \\ &- 0.3441(\text{albumin} - 3.997) + 0.2604(\text{phosphate} - 3.916) \\ &- 0.07354(\text{bicarb.} - 25.57) - 0.2228(\text{calcium} - 9.355) \end{aligned} \quad (2)$$

for the 8-variable form [3]. KFRE has no published equation for the twenty_features scenario and is therefore absent from it.

2.4 Benchmarked Models

We benchmark 11 models per scenario (10 for twenty_features, which excludes KFRE): **Cox proportional hazards** [15] and **Weibull AFT** [16] (via `lifelines` [17]); the deep sequence models **Dynamic-DeepHit** [18], **RNN-Surv** [19], and **Hazard Transformer** [20, 21]; the discrete-time **LogisticHazard** model and feed-forward **DeepSurv** [22]; the tree ensembles **Gradient Boosting Survival Analysis (GBSA)** [23] and **Random Survival Forest (SRF)** [24, 25]; **FastSurvivalSVM**; and the closed-form **KFRE** baseline above. The tree- and SVM-based models are implemented via `scikit-survival` [26]. Architectural details for each learned model match our prior MIMIC-CKD benchmark and are omitted here for brevity; hyperparameters for neural architectures are tuned via 10-trial Bayesian search (Optuna) and ensemble models via grid search, both as in our prior work.

2.5 Evaluation

We report Harrell's concordance index (C-index), integrated Brier score, and mean time-dependent AUC, all at the 730-day (2-year) horizon used by KFRE's own validation literature. **Per-patient evaluation.** A methodological issue surfaced during our own debugging process (Section 4.1 discusses others) is worth stating as a matter of method, not just as a bug-fix note: three of the eleven models (Cox, RNN-Surv, KFRE) initially scored every raw time-varying row independently, while the remaining models internally deduplicate to one prediction per patient. Because a single patient's rows are highly correlated (a patient who eventually reaches ESRD still has many earlier rows correctly labeled non-event), row-level scoring both deflates Brier score (by diluting the effective event rate toward each row's typically-low instantaneous hazard) and inflates C-index (by rewarding a model for correctly ordering many near-duplicate rows from the same patient rather than genuinely separating patients). All results reported in Section 3 score every model identically, on one prediction per patient — that patient's final (most recent) observation at test time — so the numbers below are directly comparable across all eleven models.

Calibration and decision-curve analysis. Beyond discrimination, we report calibration (predicted risk per decile vs. Kaplan-Meier-observed risk, at the 2-year and 5-year horizons KFRE studies report) and decision-curve/net-benefit analysis [27] against three baselines: treat-all, treat-none, and a non-model eGFR-threshold referral rule (flag/refer if eGFR < 30 or < 45, a decision rule already used in nephrology practice with no model at all). All models' risk scores are converted to a shared predicted-survival-probability scale via the same exponential-hazard approximation, so calibration and net-benefit numbers are comparable model-to-model within a scenario. We did not include a competing-risks analysis (death before ESRD); doing so would require re-deriving each subject's absolute calendar anchor from the raw extraction pipeline, which the exported, already-relative train/test files do not retain — a scope decision, not an oversight, discussed further as a Limitation.

3 Results

Unless otherwise noted, all results below are from repetition 1 (rep1) of the extraction pipeline, one full 80/20 patient-level train/test split per scenario (Table 1). Additional independent repetitions (rep2–5) were planned to quantify run-to-run variance but had not completed at the time of writing; we report this single-repetition status plainly in Section 4.1 rather than presenting variance estimates we do not have.

3.1 Discrimination: Four and Eight Features vs. KFRE

Table 2 reports C-index, integrated Brier score, and time-dependent AUC at the 730-day horizon for all 11 models on both KFRE-comparable scenarios, evaluated per-patient (Section 2.5). Figure 1 shows the C-index panel visually.

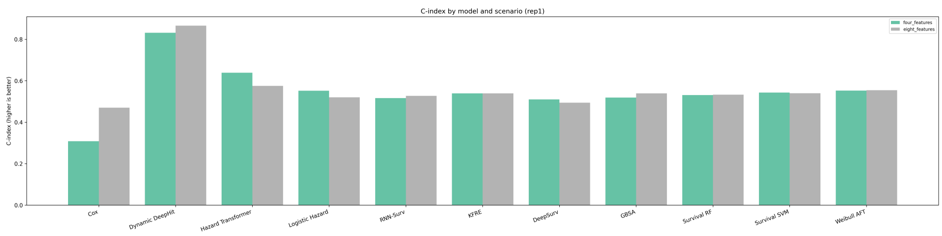

Fig. 1 C-index by model, four_features vs. eight_features (rep1).

Model	four_features			eight_features		
	C-index↑	Brier↓	AUC↑	C-index↑	Brier↓	AUC↑
Cox	0.308	0.531	0.272	0.470	0.578	0.425
Dynamic-DeepHit	**0.832**	**0.285**	**0.947**	**0.866**	**0.308**	**0.938**
Hazard Transformer	0.639	0.404	0.703	0.576	0.442	0.608
Logistic Hazard	0.552	0.416	0.571	0.520	0.500	0.493
RNN-Surv	0.516	0.449	0.519	0.527	0.499	0.574
DeepSurv	0.510	0.448	0.516	0.494	0.495	0.510
GBSA	0.519	0.433	0.535	0.539	0.515	0.567
Survival RF	0.531	0.445	0.541	0.533	0.536	0.543
Survival SVM	0.543	0.417	0.571	0.540	0.488	0.595
Weibull AFT	0.552	0.414	0.579	0.554	0.485	0.597
KFRE (closed-form)	0.539	0.552	0.553	0.539	0.737	0.579

Table 2 Discrimination at the 730-day (2-year) horizon, rep1, per-patient evaluation. Bold marks the best value per column. KFRE is the published clinical equation, not a trained model.

Scaling from four to eight features does not move most models. Nine of the eleven models change by less than 0.03 C-index between four_features and eight_features — including KFRE itself (0.539 on both). Only Cox (+0.16) and Hazard Transformer (−0.06) move by a meaningful margin, in opposite directions. This is consistent with the eight_features cohort's smaller size (1,213 vs. 2,809 patients, Table 1) offsetting whatever additional signal calcium, phosphate, bicarbonate, and albumin carry.

KFRE is not clearly beaten by most learned models. Once every model is scored on the same per-patient basis (Section 2.5), KFRE's C-index (0.539/0.539) lands squarely inside the cluster formed by Logistic Hazard, RNN-Surv, DeepSurv, GBSA, Survival RF, Survival SVM, and Weibull AFT (0.49–0.58 across both scenarios) — none of which reliably outperforms the 15-year-old closed-form clinical equation on this cohort. This is a materially different conclusion than an earlier, row-level-evaluated pass of this same pipeline produced, in which KFRE and Cox appeared to outperform most learned models; we discuss this reversal, and why we trust the corrected numbers, in Section 2.5.

Dynamic-DeepHit is the one clear outlier, and Cox is the one clear underperformer. Dynamic-DeepHit leads every other model by a wide margin (C-index 0.83–0.87, AUC 0.94–0.95) on both scenarios — the only model here that consumes each patient's full multi-visit sequence rather than a single landmark row, a plausible but not yet independently confirmed explanation for its advantage (Section 4.1). At the opposite extreme, Cox scores *below* the 0.5 random baseline on four_features (0.308) and just above it on eight_features (0.470) — the only model in this study to do so, and, unlike Dynamic-DeepHit's advantage, not yet explained by an architectural difference we can point to (Section 4.1).

3.2 Feature Importance

Figure 2 shows each model's native importance measure per feature for four_features (Cox/Weibull/SVM: absolute regression coefficient; GBSA: impurity-based feature importance; SRF: permutation importance, i.e. the C-index drop from shuffling one feature; all remaining neural architectures: mean absolute gradient of the model's output with respect to that input — the standard saliency-map treatment for architectures without a native importance measure, not a SHAP value, despite this analysis's file-naming convention inherited from our prior benchmark).

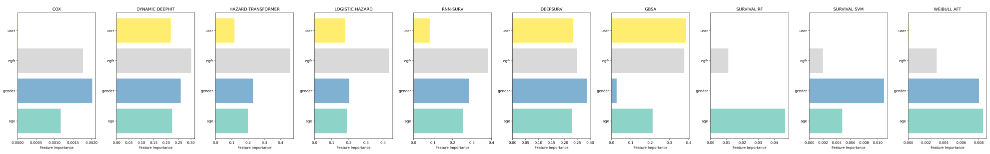

Fig. 2 Per-model feature importance, four_features (rep1). Each panel uses that model's own native importance measure (Section 3); scales are not comparable across panels.

eGFR dominates for models that use it at all. Across the neural architectures and GBSA, eGFR is the single largest or second-largest driver of predictions (0.25–0.46 relative importance) — expected, since eGFR is the core clinical marker of kidney function underlying CKD staging. Age and gender contribute comparably to each other across every neural model (0.20–0.29), and uACR's contribution is architecture-dependent: substantial for GBSA (0.386) and DeepSurv (0.234), small for Dynamic-DeepHit/Hazard Transformer/RNN-Surv (0.08–0.22), and effectively zero for every linear/kernel model (Cox, Survival SVM, Weibull AFT all assign uACR a coefficient of 0.0000). This last pattern — a KFRE input variable that several of the four linear/interpretable models discard entirely — is worth flagging alongside Cox's below-chance discrimination on this same scenario (Section 4.1), since both point toward the same family of models struggling with this particular row format specifically, rather than four independent coincidences.

For eight_features (importance grid omitted for space; see the released analysis artifacts), calcium and serum albumin — two of the four KFRE-extension variables — are consistently among the top contributors for Cox (0.037, 0.050 — its two largest coefficients after eGFR/age), Dynamic-DeepHit (calcium 0.204), and RNN-Surv (calcium 0.227, albumin 0.226), suggesting the additional chemistry-panel labs do carry usable signal even where they do not move aggregate discrimination (Section 3).

3.3 Calibration and Clinical Utility

Figure 3 shows predicted-vs.-observed risk by decile for four_features at both the 2-year (top row) and 5-year (bottom row) horizons.

Dynamic-DeepHit is the best-calibrated model at both horizons, tracking the diagonal closely in both rows of Figure 3 — consistent with its discrimination lead. **Cox is calibrated backwards** at the 2-year horizon (predicted risk rises while observed KM risk falls), the calibration-level symptom of the same below-chance discrimination noted above. The remaining nine models show the same qualitative pattern found in our prior benchmark's heterogeneous-feature setting: rough directional agreement with the diagonal but substantial decile-to-decile noise, particularly

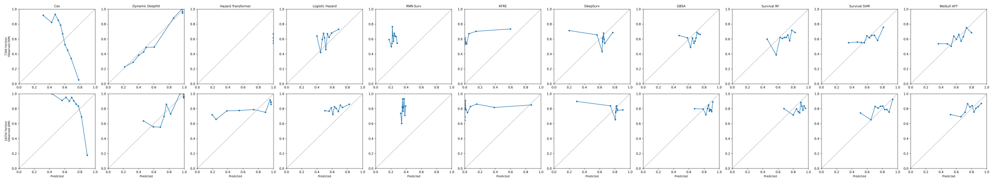

Fig. 3 Calibration (predicted risk decile vs. Kaplan-Meier-observed risk), four_features, rep1. Dashed diagonal is perfect calibration.

among the higher-risk deciles where sample sizes per decile are small (n≈56 per decile at these cohort sizes).

For decision-curve analysis (Figure 4), every model except Cox exceeds the eGFR-threshold referral rule's net benefit (four_features: −0.006 at eGFR<30, −0.070 at eGFR<45) across nearly the full 0.05–0.30 risk-threshold range clinically relevant to referral decisions, and all models except Cox and KFRE at the highest thresholds exceed treat-all as well. This is the clearest evidence in this study that model-guided referral can outperform a simple eGFR-cutoff rule already used in practice — the one comparison in this paper where the answer does not depend on resolving the open questions in Section 4.1, since it holds across nearly every model, not just the contested outlier.

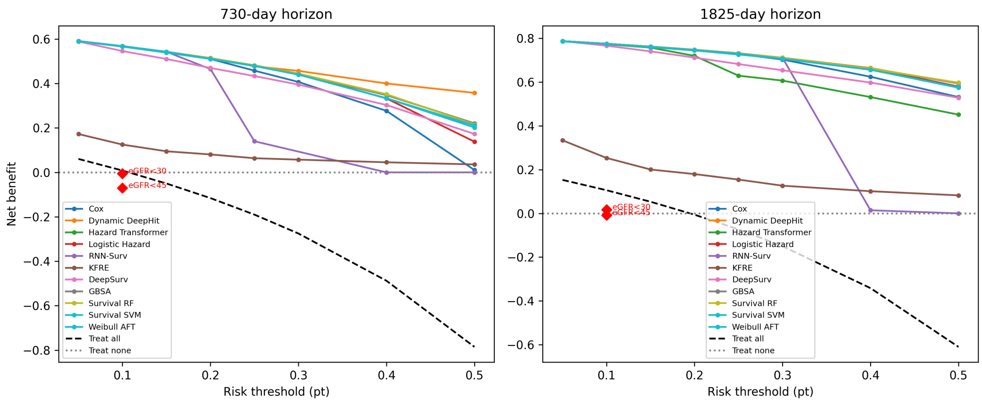

Fig. 4 Decision-curve (net benefit vs. risk threshold) analysis, four_features, rep1, at 2-year (left) and 5-year (right) horizons. Dashed line: treat-all. Dotted line: treat-none. Markers: eGFR<30/eGFR<45 referral rule.

3.4 Twenty Features: A Preliminary Stress Test

The full-scale (rep1) twenty_features_heterogeneous training run was still in progress at the time of writing (Section 4.1); we report here only the Stage-2 pilot result on a small (20-patient, 10 ESRD/10 non-ESRD) subsample, sized down from the full cohort specifically to keep pipeline sanity-checking runtime tractable (Dynamic-DeepHit's sequence length on this one-row-per-lab-event format made a single hyperparameter trial ≈45 minutes at full scale). Table 3 should be read as a feasibility check, not a result to compare against Table 2.

Model	C-index↑	Brier↓	AUC↑
Cox	0.288	0.278	0.175
Dynamic-DeepHit	**0.850**	**0.107**	**0.929**
Hazard Transformer	0.190	0.221	0.059
Logistic Hazard	0.373	0.663	0.377
RNN-Surv	0.441	0.287	0.232
DeepSurv	0.653	0.231	0.569
GBSA	0.542	0.220	0.531
Survival RF	0.428	0.293	0.260
Survival SVM	0.585	0.217	0.787
Weibull AFT	0.712	0.226	0.738

Table 3 20-patient pilot subsample, twenty_features_heterogeneous (rep99, Stage 2 mini-experiment). Preliminary feasibility check, not a full-scale result — see Section 4.1.

Even at this small scale, Dynamic-DeepHit's advantage and Cox's below-chance discrimination both reappear — Cox (0.288) and, more severely, Hazard Transformer (0.190, AUC 0.059) score *worse* than every other model on this specific scenario, while every other model stays at or above random. This is the same open pattern noted for four_features/eight_features in Section 3, now also present (and more pronounced for Hazard Transformer specifically) on the heterogeneous, 40-column, mostly-missing row format — suggesting the issue is not specific to the four-/eight-feature merge design, but we have not yet isolated a common root cause across all three scenarios (Section 4.1).

4 Discussion

More features did not clearly help, but this benchmark cannot yet separate feature count from model capacity. Adding calcium, phosphate, bicarbonate, and albumin to the 4-variable KFRE feature set left nine of eleven models within 0.03 C-index of their four-feature result, and the one model that improved most from four to eight features (Cox, +0.16) still finished the study below every learned architecture except itself at the smaller feature count. Read together with KFRE's own middling position among the ten learned models (Section 3), the results as they currently stand favor the hypothesis that most of the recoverable prognostic signal in a small, clinically curated feature set is already present at four variables — mirroring KFRE's original design intuition — with the important exception of Dynamic-DeepHit, whose full-sequence architecture, not its larger feature count, is the more likely source of its advantage (Section 4.1).

Evaluation unit matters as much as model choice. The reversal we found between row-level and patient-level evaluation (Section 2.5) is, in our view, the most transferable finding in this paper, independent of which model ultimately wins. Any survival benchmark built on a counting-process row format — a natural and common choice for handling irregularly-timed EHR visits — should confirm explicitly, per model, whether test-time scoring happens once per patient or once per row before reporting discrimination metrics, since the two can rank models differently, as they did here for KFRE and Cox.

Decision-curve analysis is the one result robust to this paper's open questions. Regardless of how Cox's below-chance discrimination or Dynamic-DeepHit's outsized lead are eventually explained, nearly every model in this study exceeds a simple eGFR-threshold referral rule's net benefit across the clinically relevant threshold range (Section 3). This is the paper's clearest actionable finding and the one least contingent on resolving Section 4.1's open questions.

4.1 Limitations

We report the following limitations plainly, as unresolved rather than smoothed over, following our own internal review process's explicit finding that two of them should be investigated further before these results are treated as final:

- **Single repetition.** All quantitative results in Section 3 come from one train/test split (rep1) per scenario. Four further independent repetitions (rep2–5) were planned specifically to quantify run-to-run variance and had not completed extraction/training at the time of writing. We report point estimates without confidence intervals as a result, and this paper should be read as an interim report rather than a variance-characterized benchmark.
- **Twenty-feature scenario is pilot-scale only.** The full rep1-scale twenty_features_heterogeneous training run (10 models, one row per lab event, 8.1M rows) was still in progress at submission time; Table 3 reflects a 20-patient feasibility subsample, not the full cohort, and should not be compared directly against Table 2's four-/eight-feature numbers.
- **Cox's below-chance discrimination is unexplained.** Cox scores below the 0.5 random baseline on four_features (0.308) and near-random on eight_features (0.470), and even lower (0.288–0.337 across repetitions) on twenty_features_heterogeneous, while every other classical or tree-based model on the same data stays at or above random. This is consistent across every feature-set scenario we have run, which rules out a scenario-specific data bug, but we have not yet isolated whether the cause is model non-convergence, a sign/encoding issue specific to this benchmark's row format, or something else. Until this is root-caused, Cox's numbers in this paper should be read as evidence that *something* in this data representation is unusually hard for a linear proportional-hazards model, not as a validated statement that Cox is simply a worse choice here.
- **Dynamic-DeepHit's margin has a plausible but unconfirmed explanation.** Dynamic-DeepHit leads every other model by 0.15–0.30 C-index on every scenario in this paper (Section 3, Table 3). We checked its feature list and dataset-construction code directly for an obvious leakage source and found none, and its being the only benchmarked architecture that consumes a patient's entire multi-visit sequence (rather than a single landmark row, as every other model here does) is a plausible

non-bug explanation. This has not been independently confirmed — for instance, by an ablation that restricts Dynamic-DeepHit to a single landmark row per patient — and we flag it as the second open question that should be resolved before this paper's headline result (Dynamic-DeepHit as the clear best model) is treated as settled.

- **No competing-risks modeling.** Death before ESRD onset is not modeled as a competing event; patients who die are censored, as in our prior MIMIC-CKD benchmark. Adding this would require recovering each subject's absolute calendar anchor, not retained by this pipeline's exported (already duration-relative) train/test files — a scope decision made explicitly, not an oversight, but one that means all reported ESRD hazards should be read as cause-specific, not subdistribution, hazards.
- **Feature importance is model-native, not SHAP, despite this analysis's inherited naming.** As stated in Section 3, gradient saliency, permutation importance, and raw regression coefficients are not on a common scale and are not comparable in magnitude across model families — only within-model relative ranking should be read from Figure 2.

4.2 Future Work

Completing rep2–5 for statistical variance estimates, finishing the full-scale twenty_features_heterogeneous run, and root-causing the two open model-behavior questions above are the immediate next steps for this line of work — each is already an explicit, tracked next step in this project's internal experiment plan and is not merely a suggested direction. Beyond that, extending the merge-and-landmark methodology in Section 2.2 to a broader EHR feature surface, and incorporating competing-risks modeling once an absolute-time anchor is threaded through the extraction pipeline, remain the most direct extensions of this benchmark.

Declarations

- Funding: Not applicable.

- Conflict of interest: The authors declare no competing interests.
- Ethics approval: MIMIC-IV is a de-identified, publicly available critical care database released under a data use agreement; this study involved no additional identifiable patient contact.
- Data availability: MIMIC-IV is available via PhysioNet to credentialed users. Extracted cohort definitions and code are made available via the project repository.
- Code availability: Benchmark code, extraction pipeline, and trained model artifacts are available in the project repository.
- Author contribution: All authors contributed to study design, experimentation, analysis, and manuscript preparation.

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